

RESCUE JUNIOR - Bill of Materials (BOM)

Team name: SERŠ TEAM

Every Line/Maze team has to submit a bill of materials for their robot

Instructions:

*Enter Team name above
*All costs need to be in local currency and their approximate conversion into dollars.
*All components worth less can be summarized in one line quantity 1 and their total cost (e.g. Screws)
*In the Local Currency Name column, the team must enter the name of the local currency, if it is different from US dollars.
*In the software sheet, in the Author column, the name of the designer, writer, or company that owns the software or library should be entered. If the team created their own algorithm, dataset, or AI model, they can include their names as authors.
*On the hardware sheet, in the "Author" column, enter the name of the company, engineer, or part manufacturer. If the team made a custom part, such as the robot housing, the team should include their names as the authors.

IMPORTANT The "Hardware" sheet is not required for simulation

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Name of the Local Currency	Total Cost Local Currency	Total Cost (\$) U.S.A. Dollars
Euro	884.98 €	\$1,026.58

#	Component	Part name	Autor	Source	Quantity	Unit cost (Local Currency)	Unit cost (USA Dollars)	Total Cost (Local Currency)	Total Cost (USA Dollars)
1	IMU	BNO055	Adafruit	https://www.adafruit.com/product/2472	1	30.13 €	\$34.95	30.13 €	\$34.95
2	PCB 160x160mm 4 Layers,Thickness:1.6 mm, Finished copper:1,Surface finish:HASL with lead	W672863ASP3	PCBWAY	pcbway.com	1	130.44 €	\$151.31	130.44 €	\$151.31
3	Raspberry camera module	Camera module 3 wide	Raspberry Pi	https://www.galagomarket.com/item/display/3308/1	2	48.74 €	\$56.54	97.48 €	\$113.08
4	Raspberry Pi 5	Raspberry Pi 5 8GB single board computer	Raspberry Pi	https://www.galagomarket.com/item/display/3746?	1	100.04 €	\$116.05	100.04 €	\$116.05
5	Battery XCell LiPo Cracker 11,1V / 2100mAh	LP 3S1P2100	XCell	https://www.delko.si/si/xcell-lipo-cracker-111v-2100mah.html	2	28.40 €	\$32.94	56.80 €	\$65.89
6	DDSM210 Motors	DDSM210	Waveshare	https://www.waveshare.com/product/robotics/moto	4	26.00 €	\$30.16	104.00 €	\$120.64
7	Programmer / debugger	ST-LINK/V2	STMICROELECTRONICS	https://si.farnell.com/stmicroelectronics/st-link-	1	21.19 €	\$24.58	21.19 €	\$24.58
8	STM32 MCU	STM32F767ZIT6	STMICROELECTRONICS	https://si.farnell.com/stmicroelectronics/stm32f767z	2	18.42 €	\$21.37	36.84 €	\$42.73
9	OLED display	MDOB128064XV-YI	Midas	https://si.farnell.com/midas/mdob128064xv-yi/oled-	2	17.46 €	\$20.25	34.92 €	\$40.51
10	Time of flight distance sensor	VL53L3CX	Pololu	https://www.pololu.com/product/3416	10	16.95 €	\$19.66	169.50 €	\$196.62
11	Top chassis 3D printed	Custom made	Tadej Božičko	-	1	10.00 €	\$11.60	10.00 €	\$11.60
12	Bottom chassis from bent aluminium	Custom made	Benko tehna	https://www.benkotehna.si/	1	10.00 €	\$11.60	10.00 €	\$11.60
13	2.3kg Serial Bus Servo, Two-way Feedback, Servo/Motor Mode Switchable, Compact Size, 300° Rotation Angle	SC09	Waveshare	2.3kg Serial Bus Servo, Two-way Feedback, Servo/Motor Mode Switchable, Compact Size, 300° Rotation Angle SC09 Servo	2	10.00 €	\$11.60	20.00 €	\$23.20
14	Sensor Board, RGB Colour, 3.3 V to 5 V, -30 °C to 70 °C, 0x29 I2C Address, Arduino Board	SEN0212	DFROBOT	https://si.farnell.com/dfrobot/sen0212/rgb-colour-sensor-board-arduino/dp/3974123	2	7.88 €	\$9.14	15.76 €	\$18.28
15	Rotary encoder	482020514001	Würth	https://si.farnell.com/wurth-	1	4.11 €	\$4.77	4.11 €	\$4.77
16	Infrared Reflective Sensor	Infrared Reflective Sensor	Waveshare	https://www.waveshare.com/infrared-reflective-	1	4.00 €	\$4.64	4.00 €	\$4.64
17	Power switch	A8L-21-12N2	Omron	https://eu.mouser.com/ProductDetail/Omron-	1	3.50 €	\$4.06	3.50 €	\$4.06
18	3V3 buck regulator	LM2576D2TR4-3.3G	On semi	https://si.farnell.com/on-	1	2.94 €	\$3.41	2.94 €	\$3.41
19	SMD Aluminium Electrolytic Capacitor, Radial Can - SMD, 10 µF, 35 V, 3000 hours @ 105°C, Polar	EXV106M035A9DAA	KEMET	https://si.farnell.com/kemet/exv106m035a9daa/cap-10-f-35v-radial-smd/dp/2069187	4	2.68 €	\$3.11	10.72 €	\$12.44
20	5V buck regulator	LM2576D2TR4-5G	On semi	https://si.farnell.com/on-	1	2.19 €	\$2.54	2.19 €	\$2.54
21	Power Inductor (SMD), 100 µH, 3 A, Unshielded	MP002808	MULTICOMP PRO	https://si.farnell.com/multicomp-pro/mp002808/power-inductor-100uh-	4	2.12 €	\$2.46	8.48 €	\$9.84
22	I2C switch	TCA9548APWR	Texas Instruments	https://si.farnell.com/texas-	2	1.68 €	\$1.95	3.36 €	\$3.90
23	USB B conn	MC32597	Muticomp	https://si.farnell.com/multicomp/mc32597/usb-2-0-	2	0.75 €	\$0.87	1.50 €	\$1.74
24	SMD Aluminium Electrolytic Capacitor, Radial Can - SMD, 1000 µF, 10 V, 2000 hours @ 105°C, Polar	MCVVTT010M102GB3L	MULTICOMP	https://si.farnell.com/multicomp/mcvvt010m102gb3l/cap-1000-f-10v-smd/dp/2611350?st=MCVVTT010M102GB3L	4	0.66 €	\$0.77	2.64 €	\$3.06
25	Schottky Rectifier, 40 V, 3 A, Single, SOD-123FA, 2 Pins, 500 mV	SS34FA	ONSEMI	https://si.farnell.com/on-semiconductor/ss34fa/schottky-rect-aec-q101-40v-	4	0.53 €	\$0.61	2.12 €	\$2.46
26	SMD Aluminium Electrolytic Capacitor, Radial Can - SMD, 100 µF, 16 V, 1000 hours @ 105°C, Polar	EEEHA1C101WP	PANASONIC	https://si.farnell.com/panasonic/eeeha1c101wp/cap-100-f-16v-radial-smd/dp/1973308?st=eeeha1c101wp	4	0.23 €	\$0.27	0.92 €	\$1.07
27	Pin Header, Top Entry, Wire-to-Board, 1.5 mm, 1 Rows, 4 Contacts, Through Hole, ZH	B4B-ZR(LF)(SN)	JST	https://si.farnell.com/jst-japan-solderless-terminals/b4b-zr-lf-sn/header-top-entry-	10	0.14 €	\$0.16	1.40 €	\$1.62
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1	STM32 Development toolchain (IDE + CubeMX)	Software's Tool	Development toolchain for developing with STM32 MCUs
2	Onshape	Software's Tool	3D design software
3	ArduPID	Library	PID controller for Arduino - modified for use with STM32
4	FireTimer	Library	Timer for helping the ArduPID library
5	TCA9548 library	Library	Arduino Library for TCA9548 I2C multiplexer - modified for use with STM32
6	SCSCL	Library	Application-layer program for the Feite SCSCL series of serial servos
7	SSD1306	Library	Library for SSD1306 display
8	BNO055	Library	Library for BNO055 IMU
9	VL53L3CX ULP	Library	Library for VL53L3CX TOF sensor
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